CHAPTER 3

"ONE SMALL STEP . . . ONE GIANT LEAP"[5]

Projects Mercury and Gemini would lead to NASA's crowning achievement in human spaceflight: Project Apollo, the remarkable U.S. space effort that sent twelve astronauts to the surface of the Earth's Moon. In any discussion of space exploration, the Moon had never been far from hand. The NASA Long Range Plan published in December 1959 had identified missions to land on the Moon as the long-range goal of the NASA human spaceflight program. Those landings were to take place "beyond 1970." Objectives for the 1965 to 1967 time period, after the completion of Project Mercury (Project Gemini had not yet been conceived), were the first launches "in a program leading to manned circumlunar flight and to [a] permanent near-earth space station."

By mid-1960, NASA's thinking about the intermediate steps in human spaceflight had matured to the point that the space agency called together representatives of the emerging space industry to share that thinking. At a "NASA-Industry Program Plans Conference" held in Washington on July 28 and 29, 1960, George Low, the head of human spaceflight at NASA's Washington headquarters, told the audience that "our present planning calls for the development and construction of an advanced manned spacecraft with sufficient flexibility to be capable of both circumlunar flight and useful earth-orbital missions. In the long range, this spacecraft should lead toward manned landings on the moon and planets, and toward a permanent manned space station. This advanced manned space flight program has been named 'Project Apollo.'" The name Apollo had been suggested by Low's boss, NASA's director for space flight programs Abe Silverstein, in early 1960.

NASA, and particularly George Low, continued to move forward in the second half of 1960 in planning what might follow Project Apollo, which at that point had only Earth orbit and circumlunar flight as its goals. On October 17, 1960, Low informed Silverstein that "it has become increasingly apparent that a preliminary program for manned lunar landings should be formulated. This is necessary in order to provide a proper justification for Apollo, and to place Apollo schedules and technical plans on a firmer foundation." To undertake this planning, Low formed a small working group of NASA Headquarters staff. The results of that group's work were soon to form the technical basis for a presidential commitment to send Americans to the Moon "before this decade is out."

That NASA was planning advanced human spaceflight missions, including ones to land people on the Moon, soon came to the attention of President Eisenhower and his advisers. The president asked his science adviser, George Kistiakowsky, to organize a study of "the goals, the missions and the costs" of the human spaceflight program that NASA had in mind. To carry out such a study, Kistiakowsky established an "Ad Hoc Committee on Man-in-Space" chaired by Brown University professor Donald Hornig. The Hornig Committee issued its report on December 16, 1960, and briefed it to President Eisenhower a few days later. Eisenhower has been quoted as saying at that time that "he couldn't care less whether a man ever reaches the Moon," and when a comparison was made to Queen Isabella's willingness to finance the voyages of Christopher Columbus, Eisenhower replied that "he was not about to hock his jewels" to send men to the Moon.

President's Science Advisory Committee, "Report of the Ad Hoc Panel on Man-in-Space," December 16, 1960

1. INTRODUCTION

We have been plunged into a race for the conquest of outer space. As a reason for this undertaking some look to the new and exciting scientific discoveries which are certain to be made. Others feel the challenge to transport man beyond frontiers he scarcely dared dream about until now. But at present the most impelling reason for our effort has been the international political situation which demands that we demonstrate our technological capabilities if we are to maintain our position of leadership. For all of these reasons we have embarked on a complex and costly adventure. It is the purpose of this report to clarify the goals, the missions and the costs of this effort in the foreseeable future, particularly with regard to the man-in-space program . . .

2. THE MAN-IN-SPACE PROGRAM

The initial American attempt to launch a manned capsule into orbital flight, Project Mercury, is already well advanced. It is a somewhat marginal effort, limited by the thrust of the Atlas booster. It has as its goal the launching of a one man capsule into orbit around the earth and its successful return to earth. The fact that the thrust of any available American booster is barely sufficient for the purpose means that it is difficult to achieve a high probability of a successful flight while also providing adequate safety for the Astronaut. Achieving reliability on both accounts will strain our capabilities. A difficult decision will soon be necessary as to when or whether a manned flight should be launched. The chief justification for pushing Project Mercury on the present time scale lies in the political desire either to be the first nation to send a man into orbit, or at least to be a close second.

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None of the boosters now planned for development are capable of landing on the moon with sufficient auxiliary equipment to return the crew safely to earth. To achieve this goal, a new program much larger than Saturn will be needed. It is likely to take one of three forms:

1. An all-chemical liquid-fueled rocket, the Nova, might be developed to take the trip directly. It would require a booster with about 6 times the thrust of the Saturn and utilizing either kerosene or hydrogen-oxygen. The upper stage of the Nova would require hydrogen-oxygen and at least one stage would probably be an existing stage from the Saturn development program.

2. If a suitable nuclear upper stage could be developed, the Nova vehicle could conceivably become a combination chemical-nuclear system. This system would still require the development of a first stage chemical booster with thrust of the same order of magnitude as that described for the all-chemical system. If the nuclear development should be as successful as its proponents hope, it might open the way for future developments beyond the possibilities envisioned for chemical rockets. However, a sound decision on the promise of nuclear rockets cannot be made until about 1963.
3. Rendezvous techniques, utilizing either Saturn C-2 vehicles or some type of advanced Saturn vehicles, could be employed to lift into an earth orbit the hardware and fuel necessary to perform the manned lunar landing mission. In this system, a series of vehicles would be launched into a temporary earth orbit where they would rendezvous to enable fueling of the spacecraft and, if necessary, assembly of the component parts of the spacecraft. This spacecraft would then be used to transport the manned payload to the moon and thence back to Earth. These techniques will require considerable development, and are at present only in a preliminary study phase.

It is clear that any of the routes to land a man on the moon require a development much more ambitious than the present Saturn program. Not only must much bigger boosters probably be developed, but rockets and guidance mechanisms for the safe landing and then for return from moon to earth by means of additional rockets must be developed and tested. Nevertheless, it must be pointed out that this new, major step is implicit in undertaking the proposed manned Saturn program, for the first really big achievement of the man-in-space program would be the lunar landing. The succeeding step, manned flight to the vicinity of Venus or Mars, represents a problem an order of magnitude greater than that involved in the manned lunar landing. Not only does it appear to be insoluble in terms of chemical rockets, thus requiring the development of suitable nuclear rockets or nuclear-powered electric propulsion devices, but it also poses serious problems in terms of life support and radiation shielding for journeys requiring times ranging from many months to years.

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4. RELATION BETWEEN MANNED AND UNMANNED SPACE EXPLORATION

Certainly among the major reasons for attending the manned exploration of space are emotional compulsions and national aspirations. These are not subjects which can be discussed on technical grounds. However, it can be asked whether the presence of a man adds to the variety or quality of the observations which can be made from unmanned vehicles, in short whether there is a scientific justification to include man in space vehicles.

It is said that an astronaut's judgment, decision-making capability and resourcefulness can increase the probability of successful accomplishment of a space mission and expand the variety and quality of observations performed. On the other hand, man's senses can be satisfactorily duplicated at remote locations by the use of available instrumentation and advances in the state of the art are continually increasing the ability to transmit information back to a central receiving point. With such an instrumented system, the decisions requiring man's mental capabilities can be performed by many men in a normal environment and with the aid of elaborate computational aids, where necessary.

The following considerations seem pertinent:

1. Information from unmanned flights is a necessary prerequisite to manned flight.
2. The degree of reliability that can be accepted in the entire mechanism is very much less for unmanned than for manned vehicles. As the systems become more complex this may make a decisive difference in what one dares to undertake at any given time.
3. From a purely scientific point of view it should be noted that unmanned flights to a given objective can be undertaken much earlier. Hence repeated observations, changes of objectives and the learning by experience are more feasible.

It seems, therefore, to us at the present time that man-in-space cannot be justified on purely scientific grounds, although more thought may show that there are situations for which this is not true. On the other hand, it may be argued that much of the motivation and drive for the scientific exploration of space is derived from the dream of man's getting into space himself.

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6. CONCLUSIONS

1. The first major goal of the man-in-space program is to orbit a man about the earth. It will cost about 350 million dollars.
2. The next goal, of an intermediate nature, is the manned circumnavigation of the moon. It will cost about 8 billion dollars.
3. The second *major* goal, landing on the moon, can only be achieved about 1975 after an additional national expenditure in the vicinity of 26 to 38 billion dollars.
4. The Saturn program is a necessary intermediate step toward manned lunar landing but must be followed by a much bigger development before manned lunar landing is possible.
5. The unmanned program is a necessary prerequisite to a manned program. Even if there were no manned program, the unmanned program might yield as much scientific knowledge and on this basis would be justified in its own right.
6. Even if there were no man-in-space program, Saturn C-2 is still a minimum vehicle for close-up instrumented study of Venus and Mars, for unmanned trips to more distant planets, and for putting roving vehicles on the surface of the moon.
7. Manned trips to the vicinity of Venus or Mars are not yet foreseeable.

When Dwight Eisenhower left office on January 20, 1961, the future of NASA's program of human spaceflight was extremely uncertain. There were no funds in Eisenhower's final budget proposal to support Project Apollo, and it was known that the incoming president, John F. Kennedy, was receiving advice skeptical of the value of launching humans into space.

As he entered the White House, Kennedy was aware that he would be faced with decisions that would shape the future of U.S. space efforts. The first order of business for the new president was to select someone to head NASA. His choice was *veteran Washington operator James E. Webb. Once Webb arrived at NASA, a first task was to review the agency's proposed budget for FY1962 that had been prepared by the outgoing Eisenhower administration. In doing so, Webb and his associates came to the conclusion that NASA's planning had been too conservative, and that the milestones included in the agency's ten-year plan should be accelerated.*

One source of input into this conclusion was the February 7 final report of George Low's Working Group, the first fully developed plan for how NASA might send men to the Moon. The report is notable for its estimate that a lunar landing program could be undertaken with the addition of $700 million per year to the NASA budget over a ten-year period. The report optimistically concluded that "no invention or breakthrough" would be needed to accomplish a lunar landing. The report recognized that NASA would need additional institutional capacity to carry out a Moon program. It also anticipated that recovery and reuse of launch vehicles would be key to lowering the costs of space exploration; that would not happen until 2017, fifty-six years in the future.

George M. Low, "A Plan for a Manned Lunar Landing," February 7, 1961

INTRODUCTION

In the past, man's scientific and technical knowledge was limited by the fact that all of his observations were made either from the earth's surface or from within the earth's atmosphere. Now man can send his measuring equipment on satellites beyond the earth's atmosphere and into space beyond the moon on lunar and planetary probes. These initial ventures into space have already greatly increased man's store of knowledge. In the future, man himself is destined to play a vital and direct role in the exploration of the moon and of the planets. In this regard, it is not easy to conceive that instruments can be devised that can effectively and reliably duplicate man's role as an explorer, a geologist, a surveyor, a photographer, a chemist, a biologist, a physicist, or any of a host of other specialists whose talents would be useful.

In all of these areas, man's judgment, his ability to observe and to reason, and his decision-making capabilities are required.

The initial step in our program for the manned exploration of space is Project Mercury. This Project is designed to put a manned satellite into an orbit more than 100 miles above the earth's surface, let it circle the earth three times, and bring it back safely. From Project Mercury we expect to learn much about how man will react to space flight, what his capabilities may be, and what should be provided in future manned spacecraft to allow man to function usefully. Such knowledge is vital before man can participate in other, more difficult, space missions.

Project Mercury is the beginning of a series of programs of ever-increasing scope and complexity . . .

The next step after Mercury is Project Apollo. The multi-manned Apollo spacecraft will provide for the development and exploitation of manned space flight technology in earth orbit; it also provide the initial step in a long-range program for the manned exploration of the moon and the planets.

In this paper we will focus on a major milestone in the program for manned exploration of space—lunar landing and exploration. This milestone might be subdivided into two phases:

1. Initial manned landing, with return to earth;
2. Manned exploration.

This report will be limited to a discussion of the initial manned lunar landing and return mission, with the clear recognition that it is a part of an integrated plan leading toward manned exploration of the moon.

An undertaking such as manned lunar landing requires a team effort on an exceedingly broad scale . . . The basic capability is provided through the parallel development of a spacecraft and a launch vehicle. Both of these developments must proceed in an orderly fashion, leading to hardware of increasing capability. Supporting these developments are many other scientific and technical programs and disciplines . . . The implementation of the manned spacecraft program requires information that will be obtained in the unmanned spacecraft and life science programs. The development of launch vehicle capability requires new engines, techniques to launch from earth orbit, and might include launch vehicle recovery developments. Both the spacecraft and the launch vehicle programs can progress only as new knowledge is obtained through advanced research . . .

NASA RESEARCH

Already there exists a large fund of basic scientific knowledge, as a result of the advanced research of the past several years, which permits confidence that the technology required for manned lunar flight can be successfully developed. It would be misleading to imply that all of the major problems are now clearly foreseen; however, there is an acute awareness of the magnitude of the problems. The present state of knowledge is such that no invention or breakthrough is believed to be required to insure the overall feasibility of safe manned lunar flight.

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It is proposed, therefore, that a vehicle larger than the Saturn C-2 be phased into the launch vehicle program in an orderly fashion following the Saturn development. Such a launch vehicle, called Nova, would use a cluster of 1,500,000 pound thrust F-1 engines in its booster stage. The exact number of F-1 engines will have to be determined later, when a more complete definition of Nova missions is in hand. Nova might be sufficiently large to permit a manned lunar landing with a single launching directly from earth. Or, although substantially larger than the Saturn C-2, it might still not be large enough to approach the moon directly from earth; in this case it would materially reduce the number of rendezvous operations needed in earth orbit for each lunar mission.

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Use of the Nova-class vehicle offers the possibility of greatly reducing the required number of launchings from earth. It might be possible to provide mission capability without rendezvous with a four-engine Nova; with an eight-engine Nova, this type of mission capability is virtually assured . . .

It is possible that other propulsion developments could contribute to manned lunar flight capability. Examples are the use of large solid propellant rockets, or nuclear propulsion. In defining a

Nova configuration, consideration will be given to both of these types of propulsion. At the present time it appears that nuclear propulsion will not be sufficiently developed for the initial manned lunar landing; however, nuclear propulsion might be very desirable and economically attractive for later exploration of the moon.

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SPACECRAFT DEVELOPMENT

The spacecraft development for the manned lunar landing mission will be an extension of the Apollo program. Before a spacecraft capable of manned circumlunar flight and lunar landing can be designed, a number of unknowns must be answered.

The two most serious questions are:

1. What are the effects on man of prolonged exposure to weightlessness?
2. How may man best be protected from radiation in space?

The entire spacecraft design, its shape and its weight, will depend to a great extent on whether or not man can tolerate prolonged periods of weightlessness. And, if it is determined that he cannot, then the required amount of artificial gravity, or perhaps of other forms of sensory stimulation, will have to be specified.

The spacecraft design and weight will also be greatly affected by the amount of radiation shielding required to protect a man. In this area, a clear definition of the pertinent types of radiation, and their effects on living beings, is needed.

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Manned landings on the moon . . . could be made in the 1968–1971 time period. If orbital operations using the Saturn C-2 vehicles prove to be practicable for this mission, then it might be accomplished toward the beginning of this range of time. On the other hand, if the spacecraft becomes much more complex than now envisioned, and consequently much heavier, a Nova vehicle will most likely be required before man can be landed on the moon. In the latter event, the program goals may not be accomplished as quickly.

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The average cost per year, over a ten year period, for the total program is of the order of $700,000,000.

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An examination of the required NASA staffing to carry out this plan was not made as a part of this study. However, it must be recognized that neither Marshall Space Flight Center nor Space Task Group, as presently staffed, could fully support these programs. If the program is to be adopted, immediate consideration must be given to this problem.

CONCLUDING REMARKS

In preparing this plan for a manned lunar landing capability, it was recognized that many foreseeable problems will require solutions before the plan can be fully implemented. Yet, an examination of ongoing NASA programs in the areas of advanced research, life sciences, spacecraft development, and engine and launch vehicle development, has shown that solutions to all of these problems should be available in the required period of time.

Throughout the plan, allowances were made for foreseeable problems; but it must be recognized that unforeseeable problems might delay the accomplishment of this mission. Nevertheless, the plan is believed to be sound in that it requires, at each point in time, a minimum commitment of funds and resources until the needed background information is in hand. Thus, the plan does not represent a "crash" program, but rather it represents a vigorous development of technology . . .

In his Inaugural Address, delivered on a wintry January afternoon, President John F. Kennedy suggested to the leaders of the Soviet Union, "together let us explore the stars." In his initial thinking, Kennedy favored using space activities as a way of increasing the peaceful interactions between the United States and its Cold War adversary. Soon after he came to the White House, Kennedy directed his science adviser to undertake an intensive review to identify areas of potential U.S.-Soviet space cooperation, and that review continued for the first three months of the

Kennedy administration, only to be overtaken by the need to respond to the Soviet launch of Yuri Gagarin on April 12, 1961. Soviet-U.S. cooperation in space was a theme that Kennedy was to return to in subsequent years.

Kennedy had not yet made up his own mind about the future of human spaceflight, and so in March he was unwilling to approve NASA's request to restore funds for the Apollo spacecraft that President Eisenhower had deleted from the NASA budget request; the sense was that decisions on this issue would come during the preparation of the FY1963 NASA budget at the end of 1961.

Events forced the president's hand much earlier than he had anticipated. In the early-morning hours of April 12, word reached the White House that the Soviet Union had successfully orbited its first cosmonaut, Yuri Gagarin, and that he had safely returned to Earth. The USSR was quick to capitalize on the propaganda impact of the Gagarin flight; Nikita Khrushchev boasted, "Let the capitalist countries catch up with our country!" In the United States, both the public and the Congress demanded a response to the Soviet achievement.

President Kennedy called a meeting of his advisers for the late afternoon of April 14 to discuss what that response might be. NASA deputy administrator Hugh Dryden told the president that to catch up with the Russians might require a crash program on the order of the Manhattan Project that developed the atomic bomb; such an effort could cost as much as $40 billion. After hearing the discussions of what might be done, Kennedy's response was "when we know more, I can decide if it's worth it or not. If someone can just tell me how to catch up. . . . There's nothing more important." Three days later, the Bay of Pigs fiasco took place, and Kennedy had added incentive to avoid looking weak on the global stage.

Kennedy had decided in December 1960 to make his vice president, Lyndon Johnson, the chairman of the existing National Aeronautics and Space Council. That council had been set up as part of the 1958 Space Act, with the president as chair. Thus legislative action was needed to give the chairmanship to the vice president. The president signed the legislation making this change on April 20, and on that same day wrote a historic memorandum to the vice president, asking him "as Chairman of the Space

Council to be in charge of making an overall survey of where we stand in space." This memorandum, written eight days after the Gagarin flight, set out very clear requirements for the U.S. response, asking for a "space program which promises dramatic results in which we could win." This was a clear signal that Kennedy intended to enter, and win, a space race with the Soviet Union.

John F. Kennedy, Memorandum for Vice President, April 20, 1961

THE WHITE HOUSE
WASHINGTON

April 20, 1961

MEMORANDUM FOR

VICE PRESIDENT

In accordance with our conversation I would like for you as Chairman of the Space Council to be in charge of making an overall survey of where we stand in space.

1. Do we have a chance of beating the Soviets by putting a laboratory in space, or by a trip around the moon, or by a rocket to land on the moon, or by a rocket to go to the moon and back with a man. Is there any other space program which promises dramatic results in which we could win?

2. How much additional would it cost?

3. Are we working 24 hours a day on existing programs. If not, why not? If not, will you make recommendations to me as to how work can be speeded up.

4. In building large boosters should we put out emphasis on nuclear, chemical or liquid fuel, or a combination of these three?

5. Are we making maximum effort? Are we achieving necessary results?

I have asked Jim Webb, Dr. Weisner, Secretary McNamara and other responsible officials to cooperate with you fully. I would appreciate a report on this at the earliest possible moment.

As he carried out his review Johnson consulted not only government agencies but also individuals whom he respected. One of those consulted was Wernher von Braun. Although von Braun as director of NASA's Marshall Space Flight Center was formally several layers down in the NASA hierarchy, he had become a publicly acclaimed space advocate, and the vice president wanted his opinions unfiltered through von Braun's NASA bosses. Von Braun met with Johnson and his associates on April 27, and followed up that meeting with an April 29 letter. His statement that the United States had an "excellent chance" of being first in sending humans to the surface of the Moon and back safely to Earth played an influential role in selecting a lunar landing as the means of winning the space race.

Letter from Wernher von Braun to the Vice President of the United States, April 29, 1961

Dear Mr. Vice President:

This is an attempt to answer some of the questions about our national space program raised by the President in his memorandum to you dated April 20, 1961. I should like to emphasize that the following comments are strictly my own and do not necessarily reflect the official position of the National Aeronautics and Space Administration in which I have the honor to serve.

Question 1. Do we have a chance of beating the Soviets by putting a laboratory in space, or by a trip around the moon, or by a rocket to land on the moon, or by a rocket to go to the moon and back with a man? Is there any other space program which promises dramatic results in which we could win?

Answer: With their recent Venus shot, the Soviets demonstrated that they have a rocket at their disposal which can place 14,000 pounds of payload in orbit. When one considers that our own one-man Mercury space capsule weighs only 3900 pounds, it becomes readily apparent that the Soviet carrier rocket should be capable of

- launching *several* astronauts into orbit simultaneously. (Such an enlarged multi-man capsule could be considered and could serve as a small "laboratory in space.")

- soft-landing a substantial payload on the moon. My estimate of the maximum soft-landed net payload weight the Soviet rocket is capable of is about 1400 pounds (one-tenth of its low orbit payload). This weight capability is *not* sufficient to include a rocket for the *return flight* to earth of a man landed on the moon. But it is entirely adequate for a powerful radio transmitter which would relay lunar data back to earth and which would be *abandoned* on the lunar surface after completion of this mission. A similar mission is planned for our "Ranger" project, which uses an Atlas-Agena B boost rocket. The "semi-hard" landed portion of the Ranger package weighs 293 pounds. Launching is scheduled for January 1962.

The existing Soviet rocket could furthermore hurl a 4000 to 5000 pound capsule *around* the moon with ensuing re-entry into the earth atmosphere. This weight allowance must be considered marginal for a one-man round-the-moon voyage. Specifically, it would not suffice to provide the capsule and its occupant with a "safe abort and return" capability, a feature which under NASA ground rules for pilot safety is considered mandatory for all manned space flight missions. One should not overlook the possibility, however, that the Soviets may substantially facilitate their task by simply waiving this requirement.

A rocket about ten times as powerful as the Soviet Venus launch rocket is required to land a man on the moon and bring him back to earth. Development of such a super rocket can be circumvented by orbital rendezvous and refueling of smaller rockets, but the development of this technique by the Soviets would not be hidden from our eyes and would undoubtedly require several years (possibly as long or even longer than the development of a large direct flight super rocket).

Summing up, it is my belief that

a. we do *not* have a good chance of beating the Soviets to a manned "*laboratory in space.*" The Russians could place it in orbit this year while we could establish a (somewhat

heavier) laboratory only after the availability of a reliable Saturn C-1 which is in 1964.

b. we *have* a sporting chance of beating the Soviets to a soft-landing of a radio *transmitter station on the moon.* It is hard to say whether this objective is on their program, but as far as the launch rocket is concerned, they could do it at any time. We plan to do it with the Atlas-Agena B-Ranger #3 in early 1962.

c. we have a sporting chance of sending a 3-man crew *around the moon* ahead of the Soviets (1965/66). However, the Soviets could conduct a round-the-moon voyage earlier if they are ready to waive certain emergency safety features and limit the voyage to one man. My estimate is that they could perform this simplified task in 1962 or 1963.

d. we have an excellent chance of beating the Soviets to the *first landing of a crew on the moon* (including return capability, of course). The reason is that a performance jump by a factor 10 over their present rockets is necessary to accomplish this feat. While today we do not have such a rocket, it is unlikely that the Soviets have it. Therefore, we would not have to enter the race toward this obvious next goal in space exploration against hopeless odds favoring the Soviets. With an all-out crash program I think we could accomplish this objective in 1967/68.

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<u>Question 5.</u> Are we making maximum effort? Are we achieving necessary results?

<u>Answer:</u> No, I do *not* think we are making maximum effort. In my opinion, the most effective steps to improve our national stature in the space field, and to speed things up would be to

- identify a few (the fewer the better) goals in our space program as objectives of highest national priority. (For example: Let's land a man on the moon in 1967 or 1968.)
- identify those elements of our present space program that would qualify as immediate contributions to this objective. (For example, soft landings of suitable

instrumentation on the moon to determine the environmental conditions man will find there.)

- put all other elements of our national space program on the "back burner."
- add another more powerful liquid fuel booster to our national launch vehicle program. The design parameters of this booster should allow a certain flexibility for desired program reorientation as more experience is gathered . . .

Summing up, I should like to say that in the space race we are competing with a determined opponent whose peacetime economy is on a wartime footing. Most of our procedures are designed for orderly, peacetime conditions. I do not believe that we can win this race unless we take at least some measures which thus far have been considered acceptable only in times of a national emergency.

On April 28, Johnson delivered an interim report to Kennedy on the results of his review of the space program. The report identified a lunar landing by 1966 or 1967 as the first dramatic space project in which the United States could beat the Soviet Union. The vice president argued that "leadership" should be the appropriate goal of U.S. efforts in space. That goal has remained influential in the years since.

Lyndon B. Johnson, Memorandum for the President, "Evaluation of Space Program," April 28, 1961

Reference is to your April 20 memorandum asking certain questions regarding this country's space program.

A detailed survey has not been completed in this time period. The examination will continue. However, what we have obtained so far from knowledgeable and responsible persons makes this summary reply possible.

Among those who have participated in our deliberations have been the Secretary and Deputy Secretary of Defense; General Schriever (AF); Admiral Hayward (Navy); Dr. von Braun (NASA); the Administrator, Deputy Administrator, and other top officials of NASA; the Special Assistant to the President on Science and

Technology; representatives of the Director of the Bureau of the Budget; and three outstanding non-Government citizens of the general public: Mr. George Brown (Brown & Root, Houston, Texas); Mr. Donald Cook (American Electric Power Service, New York, N.Y.); and Mr. Frank Stanton (Columbia Broadcasting System, New York, N.Y.).

The following general conclusions can be reported:

a. Largely due to their concentrated efforts and their earlier emphasis upon the development of large rocket engines, the Soviets are ahead of the United States in world prestige attained through impressive technological accomplishments in space.

b. The U.S. has greater resources than the USSR for attaining space leadership but has failed to make the necessary hard decisions and to marshal those resources to achieve such leadership.

c. This country should be realistic and recognize that other nations, regardless of their appreciation of our idealistic values, will tend to align themselves with the country which they believe will be the world leader—the winner in the long run. Dramatic accomplishments in space are being increasingly identified as a major indicator of world leadership.

d. The U.S. can, if it will, firm up its objectives and employ its resources with a reasonable chance of attaining world leadership in space during this decade. This will be difficult but can be made probable, even recognizing the head start of the Soviets and the likelihood that they will continue to move forward with impressive successes. In certain areas, such as communications, navigation, weather, and mapping, the U.S. can and should exploit its existing advance position.

e. If we do not make the strong effort now, the time will soon be reached when the margin of control over space and over men's minds through space accomplishments will have swung so far on the Russian side that we will not be able to catch up, let alone assume leadership.

f. Even in those areas in which the Soviets already have the capability to be first and are likely to improve upon such capability, the United States should make aggressive efforts as the technological gains as well as the international rewards are essential steps in eventually gaining leadership. The danger of long lags or outright omissions by this country is substantial in view of the possibility of great technological breakthroughs obtained from space exploration.

g. Manned exploration of the moon, for example, is not only an achievement with great propaganda value, but it is essential as an objective whether or not we are first in its accomplishment—and we may be able to be first. We cannot leapfrog such accomplishments, as they are essential sources of knowledge and experience for even greater successes in space. We cannot expect the Russians to transfer the benefits of their experiences or the advantages of their capabilities to us. We must do these things ourselves.

h. The American public should be given the facts as to how we stand in the space race, told of our determination to lead in that race, and advised of the importance of such leadership to our future.

i. More resources and more effort need to be put into our space program as soon as possible. We should move forward with a bold program, while at the same time taking every practical precaution for the safety of the persons actively participating in space flights.

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Johnson's review took place as NASA was preparing to make Alan Shepard the first American to enter space. During the same week, Kennedy asked Johnson to travel to Southeast Asia to get a sense of the situation there and determine whether direct U.S. military intervention was required. Decisions on what to recommend in response to Kennedy's April 20 memo thus needed to move quickly. Johnson wanted to get his final recommendations to the president before he left Washington on Monday, May 8; this meant that those preparing the basis for those recommendations would have to work over the weekend. The recommendations came in the form of a memorandum signed by James Webb and Secretary of Defense Robert McNamara, titled "Recommendations for Our National Space Program: Changes, Policies, Goals."

The Webb-McNamara memorandum called for an across-the-board acceleration of the U.S. space effort aimed at seeking

leadership in all areas, not only in dramatic space achievements. As its centerpiece, the report recommended getting a man to the Moon before the end of the decade. A very expensive undertaking such as sending humans to the Moon was justified, according to Webb and McNamara, on grounds of national prestige in the context of the Cold War. Johnson later that day delivered the report to the president, without modification and with his concurrence. It is, in effect, the Magna Carta of Project Apollo.

Memorandum to the Vice President from James E. Webb (NASA Administrator), and Robert S. McNamara (Secretary of Defense), May 8, 1961, with attached: "Recommendations for Our National Space Program: Changes, Policies, Goals"

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II. NATIONAL SPACE POLICY

The recommendations made in the preceding Section imply the existence of national space goals and objectives toward which these and other projects are aimed. Major goals are summarized in Section III. Such goals must be formulated in the context of a national policy with respect to undertakings in space. It is the purpose of this Section to highlight our thinking concerning the direction that such national policy needs to take and to present a backdrop against which more specific goals, objectives and detailed policies should, in our opinion, be formulated.

a. Categories of Space Projects

Projects in space may be undertaken for any one of four principal reasons. They may be aimed at gaining scientific knowledge. Some, in the future, will be of commercial or chiefly civilian value. Several current programs are of potential military value for functions such as reconnaissance and early warning. Finally, some space projects may be undertaken chiefly for reasons of national prestige.

The U.S. is not behind in the first three categories. Scientifically and militarily we are ahead. We consider our potential in